WEEK 1 · DAY 4 · OVERVIEW

Resampling and the As-Of Value



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ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

The monthly pack shows one 10-year yield per month, and the weekly sheet one per week. Tell me which observation stands for each, and write beside it the date it really comes from. No label on a day the market never had. Show me a week with no Friday. Some reports run on a holiday or a Saturday, so give me the value as of any date, and the date it came from. Then the 2026 month-end 10-year and 2s10s, tied to Excel. Two tested functions in the module, and keep any half-finished idea out of the commit.

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

Your Route Through Day 4

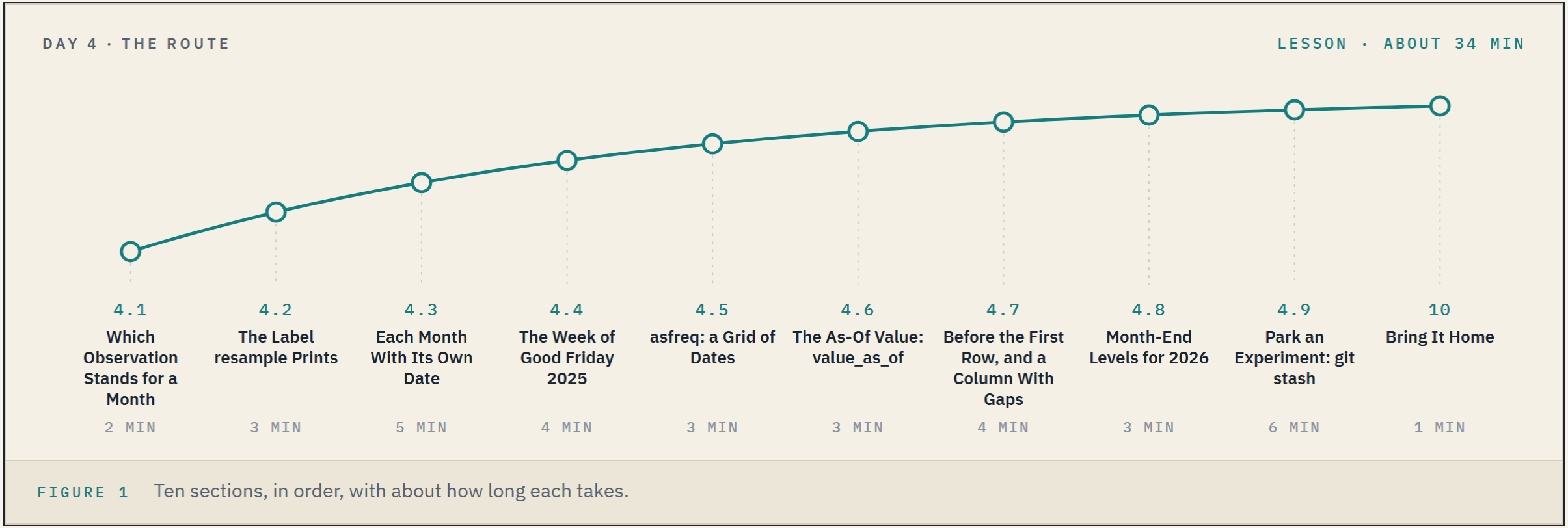
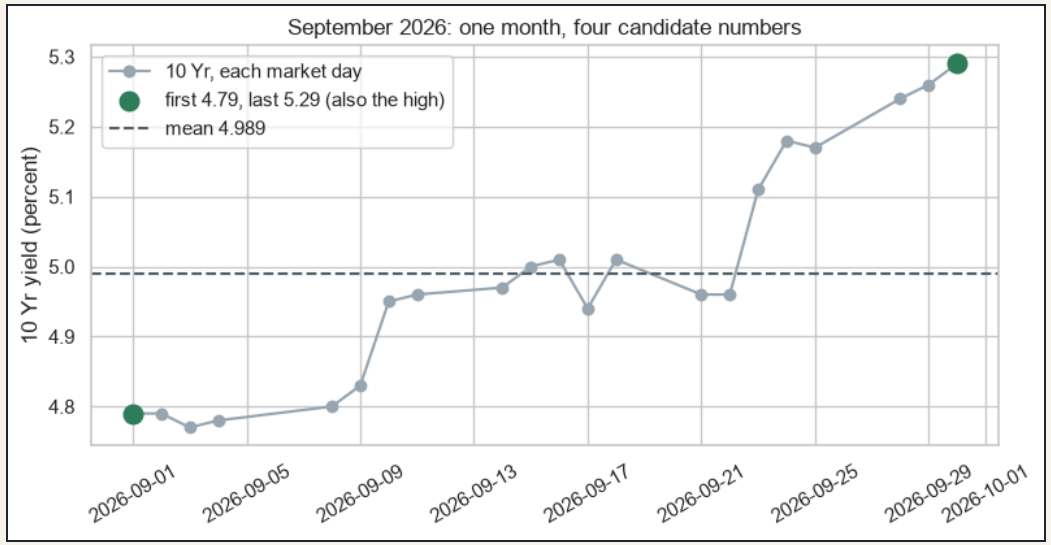


FIGURE 1 Ten sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

4.1 Which Observation Stands for a Month



EXCEL SIDE BY SIDE

JOB	EXCEL
The first row on or after 1 September	`=XLOOKUP(DATE(2026,9,1),$A$2:$A$2941,M2:M2941,,1)`

```
september = ten.loc['2026-09']
print(len(september), 'rows | first', september.iloc[0], '| last', september.iloc[-1],
      '| mean', round(september.mean(), 3), '| high', september.max())
```

```
21 rows | first 4.79 | last 5.29 | mean 4.989 | high 5.29
```

A month's number is its close: 5.29 on 2026-09-30, not the 4.989 mean or the 4.79 open.

4.2 The Label resample Prints

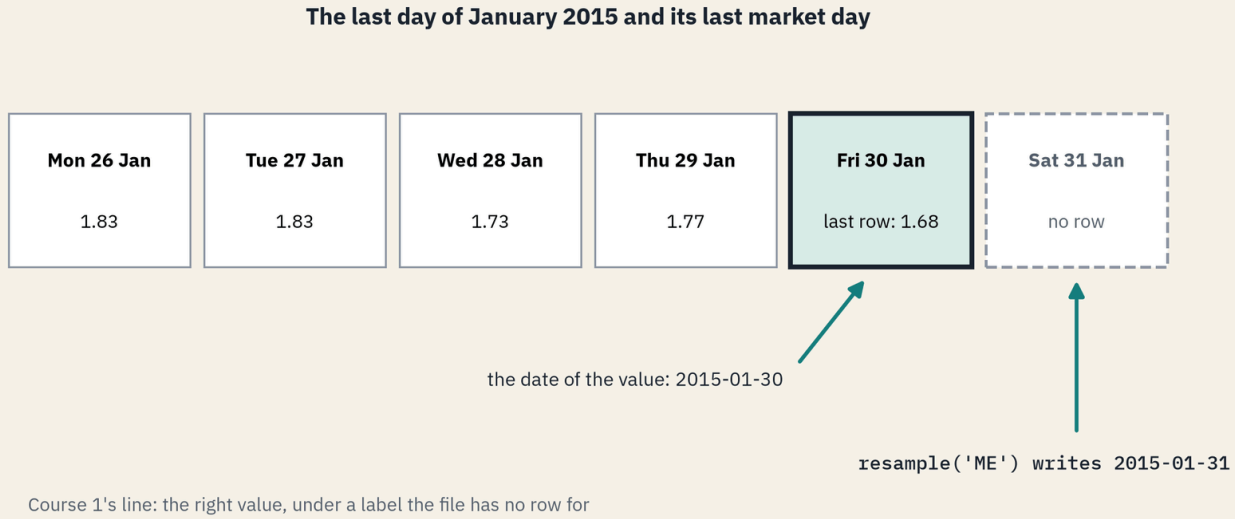


FIGURE 2 Friday 30 January 2015 holds the last row, 1.68; resample labels it Saturday 31.

COURSE 1'S LINE

```
by_resample = ten.resample('ME').last()
```

```
99 of 142 months found a 2 Yr
```

The values are right and the labels are not: a join on 43 of 142 month labels finds no row.

4.3 last_per_period: Each Month With Its Own Date

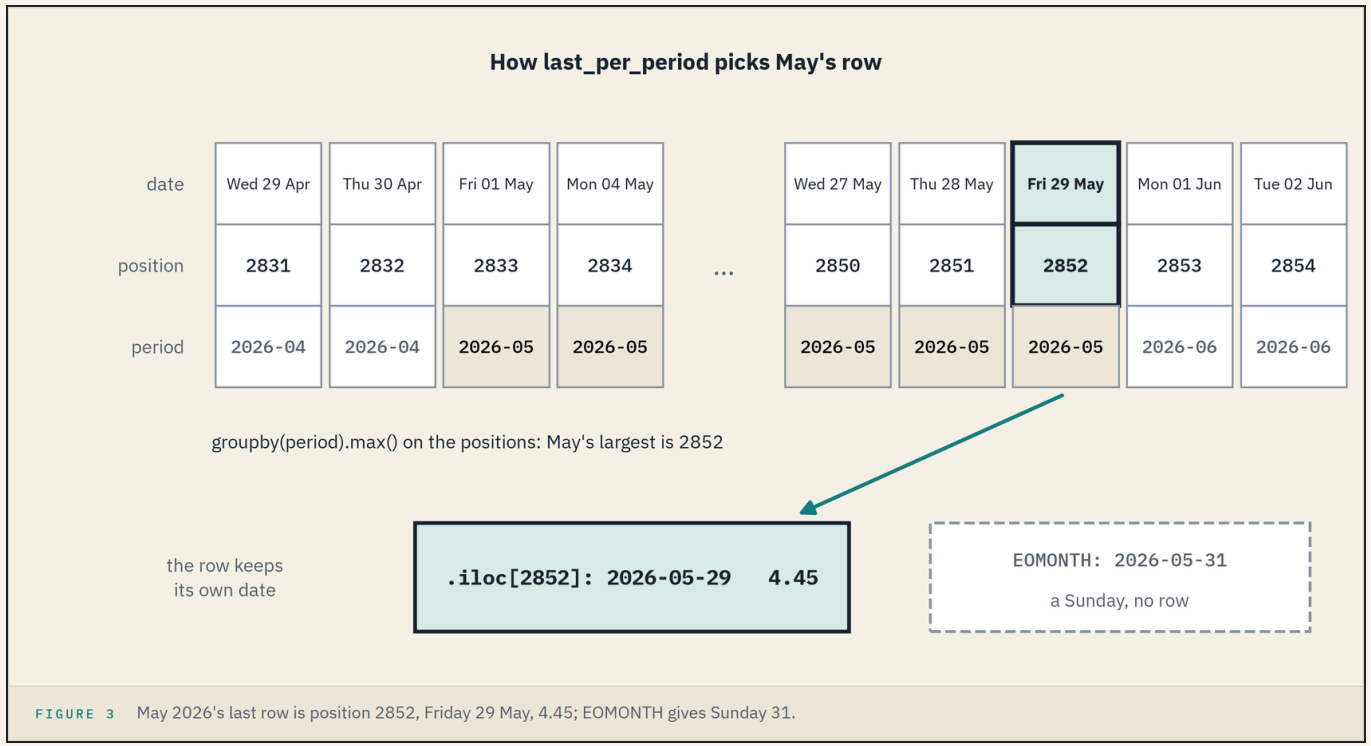


FIGURE 3 May 2026's last row is position 2852, Friday 29 May, 4.45; EOMONTH gives Sunday 31.

WHAT IT PRINTS

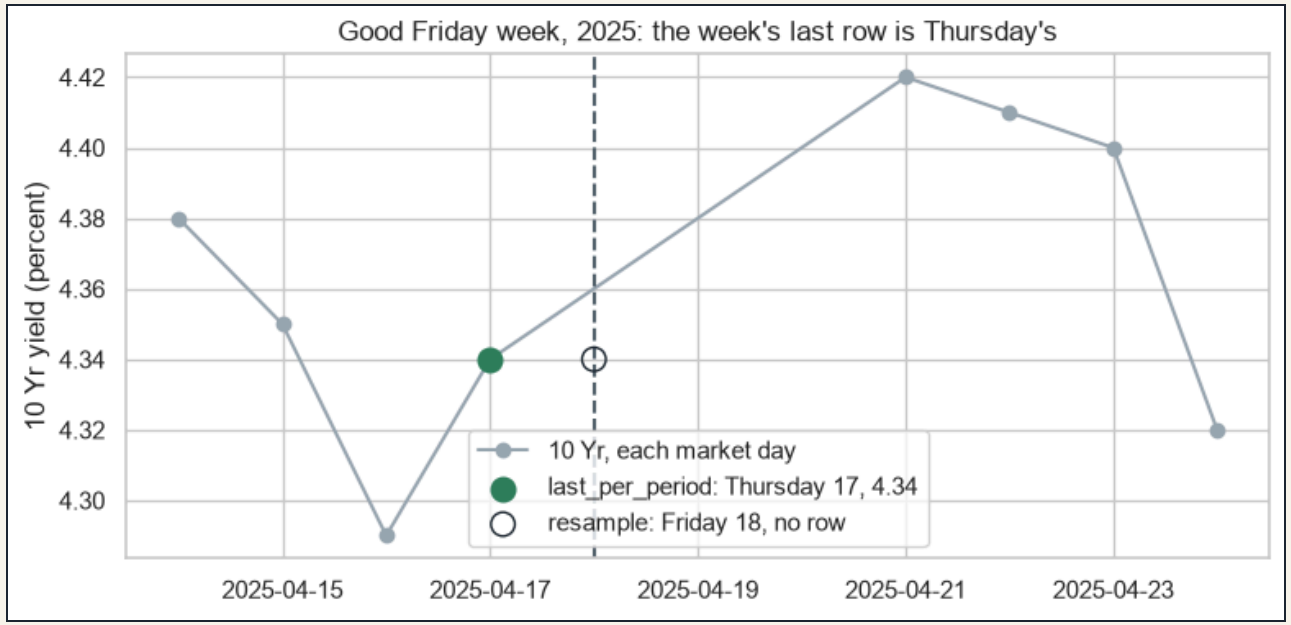
```
142 and 142 rows; values equal: True
43 labels differ; every label in the file: True
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The calendar end	`=EOMONTH(DATE(2026,5,15),0)`

`last_per_period` **keeps resample's 142 values and writes the real date: every label a market day, equal to Excel's** `MAXIFS` **then** `XLOOKUP`**.**

4.4 The Week of Good Friday 2025



BOTH LABELS

```
614 weeks
    resample last_per_period  10 Yr
0 2025-04-11      2025-04-11   4.48
1 2025-04-18      2025-04-17   4.34
2 2025-04-25      2025-04-25   4.29
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
The Friday ending a date's week	`'W-FRI'`	`=A2+MOD(6-WEEKDAY(A2),7)`	2025-04-16 gives 2025-04-18; Saturday 2025-04-19 gives 2025-04-25
That week's last market date	`around.index[1]`	`=MAXIFS($A$2:$A$2941,$A$2:$A$2941,"<="&DATE(2025,4,18))`	2025-04-17

4.5 asfreq: a Grid of Dates

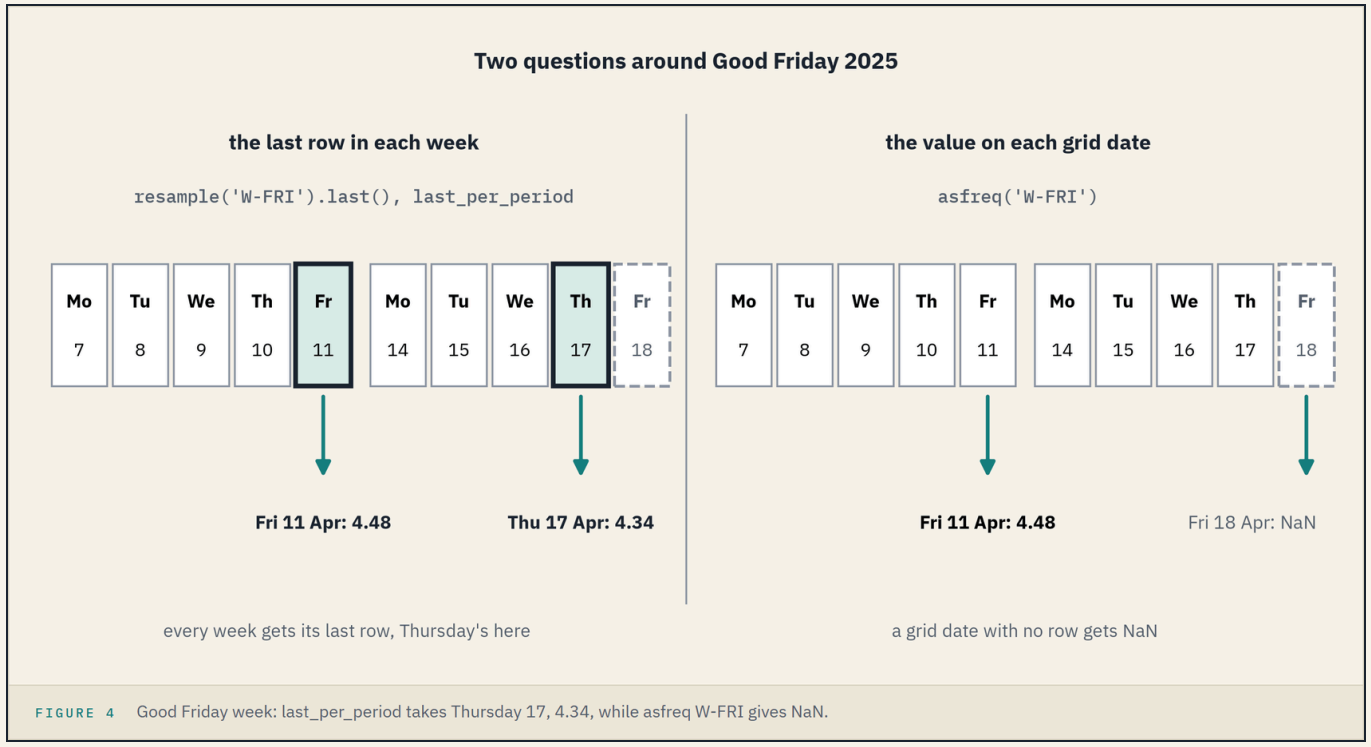


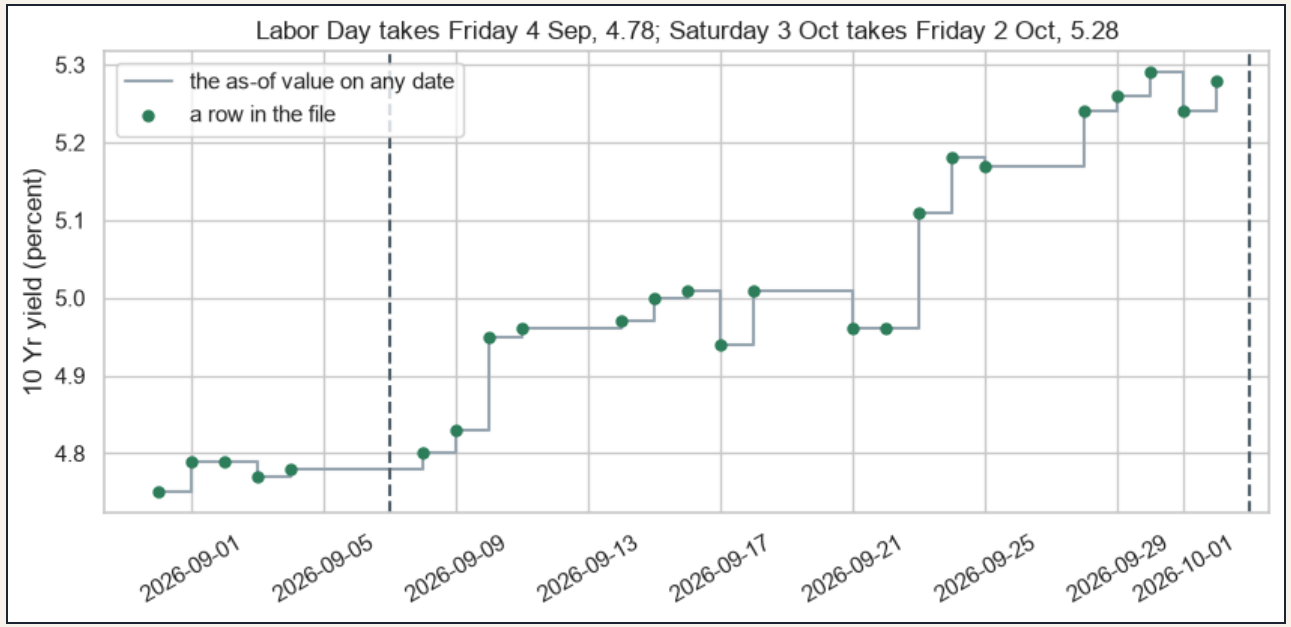
FIGURE 4 Good Friday week: last_per_period takes Thursday 17, 4.34, while asfreq W-FRI gives NaN.

EVERY WEEKDAY

```
weekday_grid = ten.asfreq('B')
holes = weekday_grid.index[weekday_grid.isna()]
```

`asfreq` asks for the value on each grid date and leaves `NaN` where the market had no row: good for finding holes, never for a level.

4.6 The As-Of Value: value_as_of



THREE DATES

```
as of 2026-09-07: 4.78 from 2026-09-04
as of 2026-10-03: 5.28 from 2026-10-02
as of 2026-09-30: 5.29 from 2026-09-30
```

EXCEL SIDE BY SIDE

JOB	EXCEL
As of Labor Day	`=XLOOKUP(DATE(2026,9,7),$A$2:$A$2941,$M$2:$M$2941,,-1)`
The date it came from	`=XLOOKUP(DATE(2026,9,7),$A$2:$A$2941,$A$2:$A$2941,,-1)`
As of Saturday 3 October	`=XLOOKUP(DATE(2026,10,3),$A$2:$A$2941,$M$2:$M$2941,,-1)`

4.7 Before the First Row, and a Column With Gaps

THE BILL WITH GAPS

```
md.value_as_of(treasury['1.5 Month'], '2026-09-07')
```

```
ValueError: series has 2532 missing values; drop them first with .dropna()
```

DROP THE EMPTY ROWS FIRST

```
print(md.value_as_of(treasury['1.5 Month'].dropna(), '2026-09-07'))
```

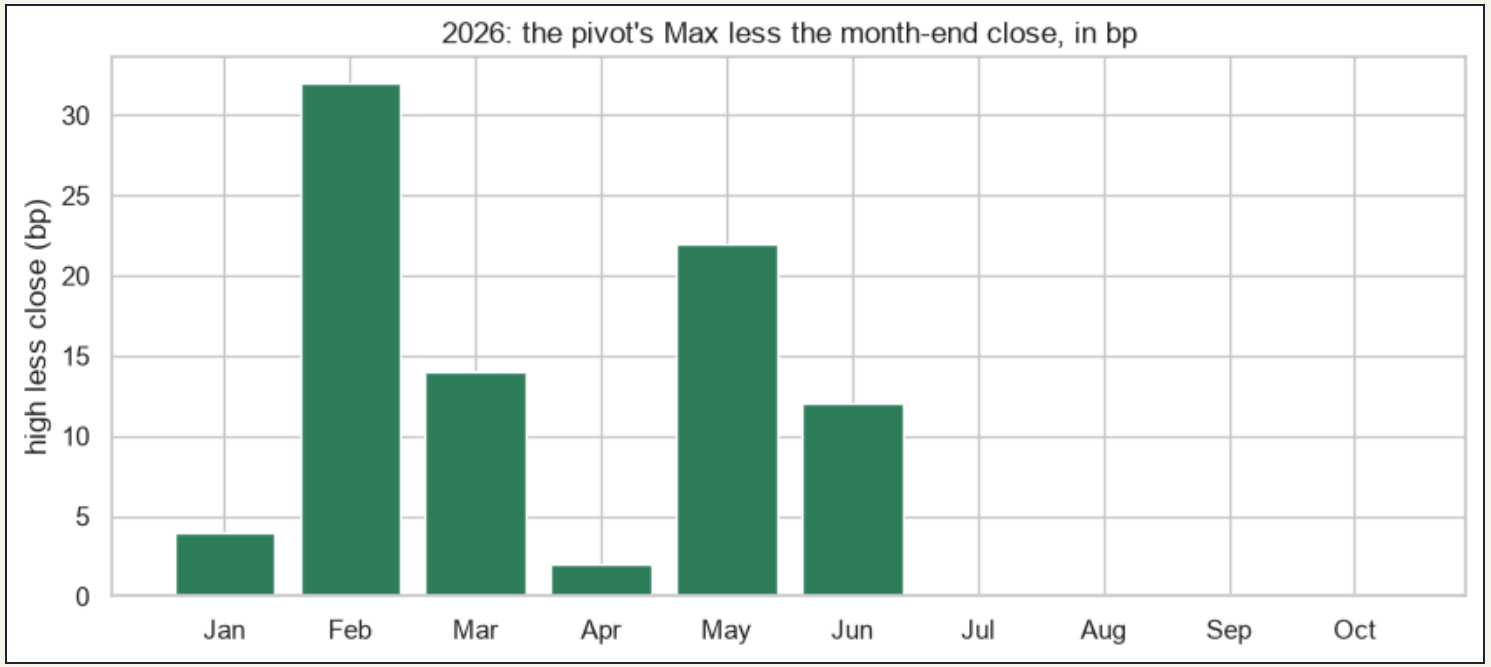
```
(Timestamp('2026-09-04 00:00:00'), 3.83)
```

> `value_as_of` never drops anything itself, so a gap is never hidden.

4.8 Month-End Levels for 2026

THE MONTH ENDS

DATE	10 YR	2S10S_BP
2026-01-30	4.26	74.0
2026-02-27	3.97	59.0
2026-03-31	4.30	51.0
2026-04-30	4.40	52.0
2026-05-29	4.45	47.0
2026-06-30	4.44	30.0
2026-07-31	4.75	47.0
2026-08-31	4.75	41.0
2026-09-30	5.29	41.0
2026-10-02	5.28	45.0



Ten month ends, each on a market day, the 10 Yr and 2s10s equal to Excel; a pivot's Max is the month's high, not its close, in 6 of 10 months.

4.9 Park an Experiment: git stash



FIGURE 5 git stash moves month_first to the shelf and hands back the tested commit; pop brings it back.

TAKE IT BACK

```
!git -C "{work}" stash pop -q
!git -C "{work}" stash list
!git -C "{work}" diff --stat
```

```
marketdata.py | 6 ++++++
 1 file changed, 6 insertions(+)
```

```
month_first in the module: False
```

`git stash` **parks half-done work and hands back the committed file;** pop **returns it, and nothing half-done reaches a commit.**

Copy the tested module into your own folder.

Bring It Home

COPY THE MODULE HOME

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_04_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py" .
# the Treasury file: read by the tests from today, never committed
Copy-Item "$work\treasury_par_yields.csv" .
python -m pytest -q test_marketdata.py
git status --short
git add marketdata.py test_marketdata.py
git commit -m "Notebook 04: last_per_period and value_as_of"
```

pytest shows 17 `passed`, and `git status` lists the Treasury file as `??`: it stays out of the commit.

You Can Now

- Say which observation stands for a month or a week, and why the course takes the close, not the mean.
- Read **`resample`**'s period-end labels, and find the ones the file has no row for.
- Take each period's last row with its own date, **`last_per_period`**, tied to **`MAXIFS`** and **`XLOOKUP`**.
- Show a week with no Friday, labelled by its Thursday.
- Use **`asfreq`** to find holes on a calendar grid, never for a level.
- Give the value as of any date, with the date it came from, **`value_as_of`**, tied to **`XLOOKUP`** with -1.
- Park half-done work with **`git stash`**, and bring it back with **`git stash pop`**.

THE NOTE ASKED	ANSWER
Which observation stands for a month or a week	The last: 5.29 on 2026-09-30 for September
The date beside it	**`last_per_period`**: 142 months, every label a market day (43 of resample's are not)

NEXT
WEEK 1 · DAY 5

Case Study: The Market Dataset

A cache that gives back every digit and refuses licensed data, then the daily dataset weeks 2 and 3 start from, checked clean.

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